LQG Regulator for Control Moment Gyroscope based Balancing System

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***Abstract*—Control Moment Gyroscope (CMG) is used extensively in many mechanical balancing systems because of its power-efficient operation. This study aims to explore the balancing capability of CMG when using the Linear–Quadratic–Gaussian (LQG) regulator. A two rigid body model physical system including a CMG and a single degree of freedom balancing object is modeled. LQG-based balancing control system with Kalman filter for estimating the state of the balancing object is developed. Simulation results show that the LQG regulator achieves better performance than the conventional PID one in terms of reaction time, noise and weight. An experimental setup shows that the CMG is able to balance a heavy system with a small tilt angle. The system can even keep balance when an external force is applied thanks to the counter torque provided by the CMG.**

I. INTRODUCTION

Control moment gyroscope (CMG) plays an important role in many applications that require balancing such as rocket, satellite and ship [1], [2]. Due to its capability of producing a large angular momentum with low power consumption, CMG is preferred to reaction wheel when the balancing object is heavy.

Two essential parts of a CMG is a flywheel for generating angular momentum and a motor for changing the flywheel rotational axis, a.k.a., gimbal angle, such that the system produces a torque toward intended direction. This torque can be pre-calculated to counteract with the moment generated by an external force applied to the system, thereby making the system balanced [3], [4], [5]. Therefore, CMGs are often embedded with complex control algorithms to fully utilize the gyroscopic effect. A self-balancing bicycle design can be found in [3] where the bicycle is kept vertical at zero-forward speed using the structure-specified mixed H2/H∞ controller. In [4], the first order sliding mode controller also exhibits the excellent capability of balancing a bike even when the bike is tilted by a large angle. Both of the algorithms show the advantages over the conventional PID control algorithm for single-input single-output (SISO) system.

One of the fundamental characteristics of CMG is that, if the angle is rotated beyond a certain threshold, the gyroscopic effect disappears [6]. Therefore, the balancing capability is reduced as the gimbal axis rotates far from the equilibrium position. In [3], the long-term balancing capability is not taken into account as the flywheel rotation axis is left uncontrolled. Other works [4], [7] consider algorithms to recover the axis to its initial value. However, the recovering time is long.

Linear–Quadratic–Gaussian regulator is a traditional control algorithm suitable for nonlinear systems such as CMG [5], [8]. One of its advantages is the capability of estimating the system state by means of Kalman filtering [9]. This study focuses on designing a control algorithm for CMG to balance a single degree-of-freedom balancing object. By applying the LQG regulator for controlling the precession of the gyroscope, the object is kept balanced with the small tilt angle of error. The advantages of the LQG regulator is exhibited when its performance is compared with PID one. Beside balancing task, the LQG takes the gimbal into account and can recover it in short time.

The rest of this paper is as follows. Section II presents the dynamic analysis of single degree of freedom balancing system using Lagrange mechanics, linearizing the dynamic equation. Section III provides the features of LQG regulator and the PID controller for SIMO systems. Section IV presents the simulation and experimental results and section V concludes the work.

II. SYSTEM MODEL

The balancing system consists of a CMG and a static object need to be balanced. Figure 1 shows the front view and the side view of a zero forward speed bicycle as the balancing object. As such, the system can be understood as an inverted pendulum. In bicycle model, the CMG is mounted on the bicycle frame. As the bicycle has the centre of gravity higher than the ground, it tends to fall down. The bicycle is considered to be balanced if the tilt angle oscillates around the equilibrium point. The tilt angle θ represents the state of the bicycle during the falling down. If the flywheel in CMG rotates with a sufficiently high speed ω, the CMG can generate a torque to counteract the falling by changing the gimbal angle α.

The structure of a CMG is shown in Figure 2. It consists of a flywheel, a frame for hanging the flywheel, called gimbal, a motor to control the flywheel rotation angle with respect to the frame, a.k.a., gimbal angle, and the circuitry system for sensing and controlling the system behavior. During system's operation, the flywheel rotates at high speed and its rotation axis is controlled by the gimbal motor. The controller measures the tilt angle and the gimbal angle to control the gimbal motor making the CMG generating a torque. Because the CMG is

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mounted on the bicycle, this torque can counteract the moment of the bicycle gravity and helps the bicycle balance.

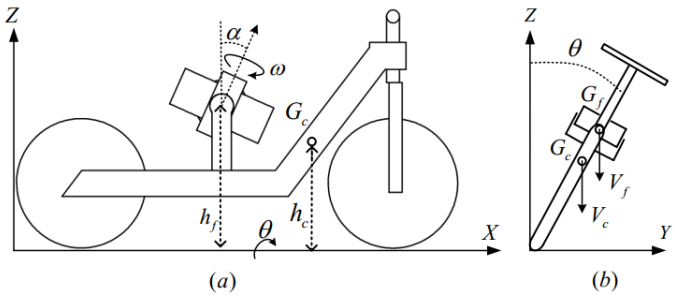

Fig. 1. The physical effect on bicycle model with a single CMG: (a) side view; (b) front view

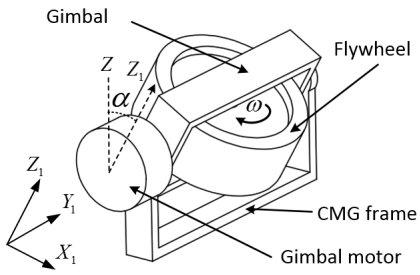

Fig. 2. The Control Moment Gyroscope (CMG) model

For modeling the gyroscopic effect and the controlling algorithm, the physical system is divided into two rigid bodies: the high-speed rotary flywheel with weight m_f and the chassis of all other static components: the bicycle, the gimbal, the gimbal motor and other electronic circuitry. The total weight of the chassis is m_c.

The nonlinear dynamics of these two rigid bodies derived by using Lagrange equations [10]:

$$\frac{d}{dt}\left\{\frac{\partial T}{\partial \dot{q}_i}\right\} - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} = Q_i^*, \tag{1}$$

where T, V, Q_i^* and q_i is total kinetic energy, total potential energy, virtual work and generalized coordinate, respectively. In this work, θ and α are chosen as the generalized coordinates. The virtual works of this model are the torques applied to the corresponding axes [11]. The kinetic and potential energy equations are derived for the bicycle chassis and the flywheel represented by the following calculations.

The total potential energy of this bicycle model is:

$$V = V_c + V_f, \tag{2}$$

where V_c and V_f are the potential energy of the chassis and the flywheel, respectively, and are calculated as:

$$V_c = m_c h_c g \cos\theta, \quad V_f = m_f h_f g \cos\theta, \tag{3}$$

where g, h_c and h_f are the gravitational acceleration, the heigh of the bicycle's center of mass, and the height of the flywheel's center of mass, respectively. The total kinetic energy of each extended object can be expressed as the sum of the translational kinetic energy of the centre of mass (TKE) and the rotational kinetic energy about the centre of mass (RKE) [12]. The translational velocities of centre of mass for the bicycle chassis and the flywheel are given by:

$$|v_c| = \dot{\theta} h_c \quad \text{and} \quad |v_f| = \dot{\theta} h_f, \tag{4}$$

respectively, The total kinetic energy of bicycle includes two components

$$T = T_c + T_f. \tag{5}$$

where T_c is the total kinetic energy of the bicycle chassis and T_f is the total kinetic energy of the flywheel. Based on the physical model in Figure 1, the total kinetic energy of the bicycle chassis is:

$$T_c = \frac{1}{2} m_c |v_c|^2 + \frac{1}{2} I_c \dot{\theta}^2, \tag{6}$$

where I_c is the moment of inertia of the bicycle chassis. Let three coordinate axes X_1, Y_1 and Z_1 are associated with the CMG as in Figure 2. The kinetic energy of the flywheel is determined by the flywheel moment of inertia I_r around X_1 axis and the flywheel polar moment of inertia I_p around Z_1 axis, and is calculated as:

$$\begin{aligned} T_f = &\frac{1}{2} m_f |v_f|^2 + \frac{1}{2} I_r \left(\dot{\theta}\cos\alpha\right)^2 \\ &+ \frac{1}{2} I_p \left(\dot{\theta}\sin\alpha\right)^2 + \frac{1}{2} I_r \dot{\alpha}^2 + \frac{1}{2} I_p \omega^2, \end{aligned} \tag{7}$$

where four terms are TKE around X axis, RKE around X axis, RKE around Y_1 axis and Z_1 axis of flywheel, respectively. From (6) and (7), total kinetic energy of the system is:

$$\begin{aligned} T = &\frac{1}{2} m_c \dot{\theta}^2 h_c^2 + \frac{1}{2} I_c \dot{\theta}^2 + \frac{1}{2} m_f \dot{\theta}^2 h_f^2 + \frac{1}{2} I_r \dot{\alpha}^2 \\ &+ \frac{1}{2} I_r \left(\dot{\theta}\cos\alpha\right)^2 + \frac{1}{2} I_p \left(\dot{\theta}\sin\alpha\right)^2 + \frac{1}{2} I_p \omega^2. \end{aligned} \tag{8}$$

Replacing (2), (8) to (1) with θ and α as the generalized coordinate, and following [3], after some calculations, two Lagrange equations are shown as:

$$\begin{aligned} &\ddot{\theta}\left(m_c h_c^2 + m_f h_f^2 + I_r \cos^2\alpha + I_p \sin^2\alpha + I_c\right) \\ &+ 2\sin\alpha\cos\alpha\left(I_p - I_r\right)\dot{\theta}\dot{\alpha} - \left(m_c h_c + m_f h_f\right) g \sin\theta \\ &= I_p \omega \dot{\alpha} \cos\alpha, \end{aligned} \tag{9}$$

$$I_r \ddot{\alpha} - \dot{\theta}^2 \left(I_p - I_r\right) \sin\alpha\cos\alpha = T_s - I_p \omega \dot{\theta} \cos\alpha - B_m \dot{\alpha}, \tag{10}$$

where T_s is generated torque and B_m are the viscosity coefficient of DC motor.

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The torque generated by DC motor is [13]:

$$T_s = K_g K_t i_m, \tag{11}$$

$$U = L\frac{di_m}{dt} + Ri_m + K_v\dot{\alpha}, \tag{12}$$

where K_g is the ratio chain transmission of gimbal motor, R and L are resistance and inductance of the motor, K_t and K_v are torque and back EMF constants of DC motor, i_m and U are supply current and supply voltage for DC motor, respectively.

From (9), (10) and (12), the three differential equations describing the dynamic model of self-balancing bicycle are shown as follows:

$$\begin{aligned}\ddot{\theta} = &\frac{I_p\omega\dot{\alpha}\cos\alpha - 2\sin\alpha\cos\alpha\left(I_p - I_r\right)\dot{\theta}\dot{\alpha}}{m_c h_c^2 + m_f h_f^2 + I_r\cos^2\alpha + I_p\sin^2\alpha + I_c} \\ &+ \frac{\left(m_c h_c + m_f h_f\right) g\sin\theta}{m_c h_c^2 + m_f h_f^2 + I_r\cos^2\alpha + I_p\sin^2\alpha + I_c},\end{aligned} \tag{13}$$

$$\ddot{\alpha} = \frac{K_g K_t i_m - I_p\omega\dot{\theta}\cos\alpha - B_m\dot{\alpha} + \dot{\theta}^2\left(I_p - I_r\right)\sin\alpha\cos\alpha}{I_r}, \tag{14}$$

$$\dot{i}_m = \frac{U - K_v\dot{\alpha} - Ri_m}{L}. \tag{15}$$

By linearizing (13), (14) and (15) around equilibrium point, i.e., $\theta = 0°$, define $x = \begin{bmatrix}\theta & \dot{\theta} & \alpha & \dot{\alpha} & i_m\end{bmatrix}^T$, $y = \begin{bmatrix}\theta & \alpha\end{bmatrix}^T$ and $u = U$. The state-space equation describes dynamic model of auto-balancing bicycle is shown as:

$$\begin{cases}\dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t)\end{cases}, \tag{16}$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ \frac{g(m_c h_c + m_f h_f)}{I_1} & 0 & 0 & \frac{I_p\omega}{I_1} & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & -\frac{I_p\omega}{I_r} & 0 & -\frac{B_m}{I_r} & \frac{K_g K_t}{I_r} \\ 0 & 0 & 0 & -\frac{K_v}{L} & -\frac{R}{L} \end{bmatrix}, \tag{17}$$

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{L}\end{bmatrix}^T, \tag{18}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \tag{19}$$

$$D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tag{20}$$

$$I_1 = m_c h_c^2 + m_f h_f^2 + I_r + I_c. \tag{21}$$

The nonlinear model of auto-balancing system is constructed in simulation environment base on three equations (13), (14) and (15). From the parameters of simulation and the experiment, the model in (16) is used for calculating the gain matrices of the LQG regulator.

III. CONTROLLER DESIGN

In order to optimize the performance of the CMG for an object in the balancing process, an optimal controller is applied to the system model. This study uses a Linear-Quadratic-Gaussian regulator because of its advantage over conventional controllers. The state space equations of the open loop plant for applying LQG regulator has the form [9]:

$$\begin{cases}\dot{x}(t) = Ax(t) + Bu(t) + v(t) \\ y(t) = Cx(t) + Du(t) + w(t)\end{cases}, \tag{22}$$

where $x(t)$, $y(t)$, $u(t)$, $v(t)$ and $w(t)$ are state vector, measured output vector, control input vector, disturbance affects this system and the measurement noise, respectively, where the disturbance and measurement noise follow Gaussian distribution. This model needs to be controlled by feedback gain matrix K, and it requires all state variables are measured. In fact, not all state variables are measurable. Therefore, a state observer using Kalman filter is added to the control system. Beside estimating the non-measureable parameters, the state observer filters the measured signals namely theta and alpha. The state-space equations of Kalman filter are given by:

$$\begin{cases}\dot{\hat{x}}(t) = (A - LC)\hat{x}(t) + Bu(t) + Ly(t) \\ u(t) = -K\hat{x}(t)\end{cases}, \tag{23}$$

from the control input $u(t)$ and the system output $y(t)$, the estimated states $\hat{x}(t)$, Kalman filter can estimate all state variables. Then, it provides all state variable value to the controller for calculating the control signal. The combination of Kalman Filter and feedback gain matrix K connect with the controlled subject creating a closed-loop system which using estimated state $\hat{x}(t)$ as measured states $x(t)$ and shows in Figure 3.

The feedback gain matrix K and Kalman gain matrix L are calculated by solving the Algebraic Riccati Equation, where the expenditure of the energy of the control signal is minimized [14].

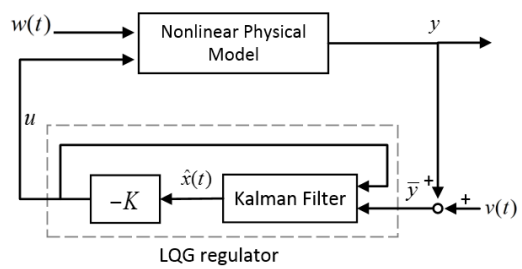

Fig. 3. The LQG feedback configuration

The control voltage for DC motor $u(t)$ is calculated base on output $y(t)$ and the previous control signal. The nonlinear physical model is constructed from equations (13), (14), and (15). In order to verify the performance of the LQG controller,

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a PID controller was also applied. The structure of PID controller is shown in Figure 4 [15].

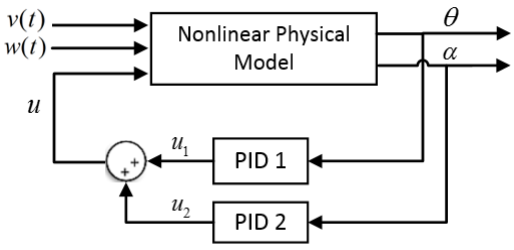

Fig. 4. The PID controller configuration

This control model constructs by two PID controllers. These controllers calculate separately two control signals (u_1 and u_2) for one general input of bicycle model. The PID controller has the form:

$$u_i = K_P e_i(t) + K_D \frac{d}{dt} e_i(t) + K_I \int e_i(t')dt', \qquad (24)$$

where K_P, K_D and K_I are proportional, derivative and integral gains, respectively. And $e_i(t)$ is the difference between the desired angle value $r(t)$ and the measured angle value $y(t)$. In this model, $e_1(t) = \theta(t)$ and $e_2(t) = \alpha(t)$.

IV. RESULTS

The performance of the CMG is evaluated by simulating the CMG model with LQG and PID controllers in different conditions.. The CMG also assessed by implementing an experiment where an inverted pendulum is controlled by LQG regulator.

A. Simulation results

For simulation, the parameters of the CMG are listed in Table I.

TABLE I
PARAMETER VALUES FOR SIMULATION

Symbol	Value	Unit
$m_c,\ m_f$	3.04, 0.61	kg
$h_c,\ h_f$	0.096, 0.11	m
ω	15000	rpm
g	9.8	m/s^2
$I_c,\ I_p,\ I_r$	0.04375, 0.00034, 0.000225	$kg.m^2$
L	0.67	H
R	18.45	Ω
$B,\ K_g,\ K_w$	0.12, 56, 5	$times$
$K_t,\ K_v$	0.0215, 0.0206	Vs/rad

Base on these parameters, the feedback gain matrix K and the estimation gain matrix L for LQG regulator are calculated as in [15]:

$$K = \begin{bmatrix} 2021.7 & 160.2 & -1000 & 5.33 & 173.6 \end{bmatrix},$$

$$L = \begin{bmatrix} 9.83 & -2.52 \\ 1.45 & 0.59 \\ -2.52 & 11.20 \\ -53.47 & 15.88 \\ 0.14 & 0.24 \end{bmatrix}.$$

In this simulation, the equilibrium point of the CMG is set at $0°$. Therefore, the system is considered to be balanced if the tilt angle is varying below $3°$ during simulation.

Figure 5 shows the behavior of the system in terms of tilt angle θ and gimbal angle α over time with the initial conditions $\theta = -25°$, $\alpha = -20°$, $\dot{\theta} = 0$, $\dot{\alpha} = 0$, $i = 0$. As initial, the system is tilted, the gimbal rotates from $-20°$ to $25°$ generating a torque to rotate the inverted pendulum to its vertical position ($\theta = 0°$) in 1 second. After that, the system is kept balanced with a small tilt angle due to the measurement noise. In this case, the overshoot of the gimbal angle is $25°$, the overshoot of tilt angle is only $5°$ and the equilibrium is reached only after one oscillation of θ and α.

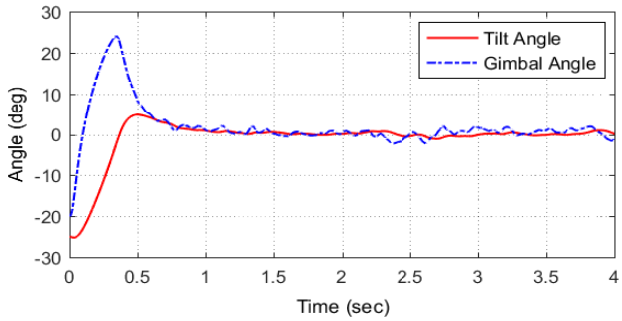

Fig. 5. System response with LQG regulator

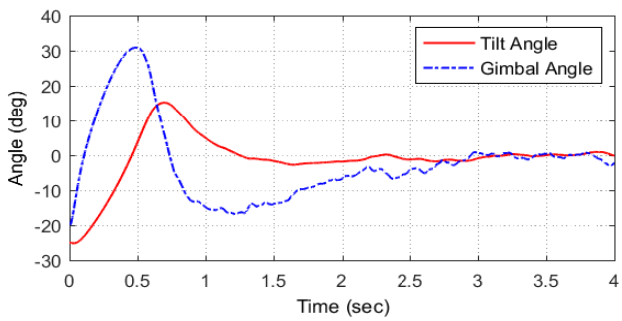

Fig. 6. System response by PID control

Figure 6 shows the simulation result by using PID controller. The parameters of PID controller are selected by changing the gain values to achieve the smallest stabilizing time. As such, the resulting PID parameters are:

$$K_{P1} = -96.5, \quad K_{D1} = 0.01, \quad K_{I1} = -79.3,$$
$$K_{P2} = 22.8, \quad K_{D2} = -4.1, \quad K_{I2} = 27.4.$$

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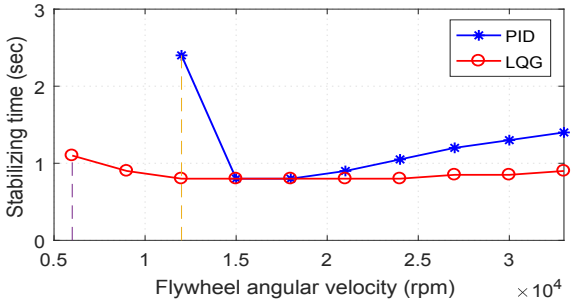

Fig. 7. The stabilizing time when changing flywheel speed

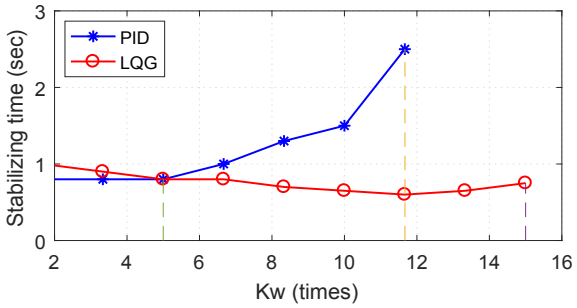

Fig. 8. The stabilizing time of system when changing weight scale

It is observed that the PID needs more than 2 seconds to balance the system, the equilibrium is reached after two oscillations of α. We can see that, LQG regulator can stabilize the system faster and the overshoots of two angles are smaller.

To test the capability of the CMG, in the next simulation, the performance in terms of stabilizing time is evaluated depending on the CMG parameters. For each CMG parameter set, LQG and PID parameters are calculated and selected for the best performance. Figure 7 shows the stabilizing time of the CMG driven by two controllers at different angular velocities of the flywheel. The system controlled by LQG reaches stability in less than 1 second for a wider range of velocity. On the other hand, PID has the only equivalent performance to LQG at the velocity of 15000 and 18000 rpm. If the flywheel is rotating too slowly, the PID can not keep the system stable.

Figure 8 shows the stabilizing time of the system when system weight varies. Let the ratio K_w of chassis weight and flywheel weight define as $K_w = m_c/m_f$. In applications, this coefficient denotes the ability of gyroscope to balance a system that whose weight is K_w times heavier than the weight of flywheel. When the weight coefficient K_w raises, the stabilizing time of PID increases while the response time of LQG remains below 1 second. The stabilizing time of LQG is always less than one second while the stabilizing time of PID is larger that one second when K_w reach 6. The highest K_w that PID keeps the system stable is only 12. Whereas, the LQG offers much better performance with K_w is 15.

In the last simulation, the robustness of the two controllers are compared. Figure 9 show the standard deviation (SD) of tilt

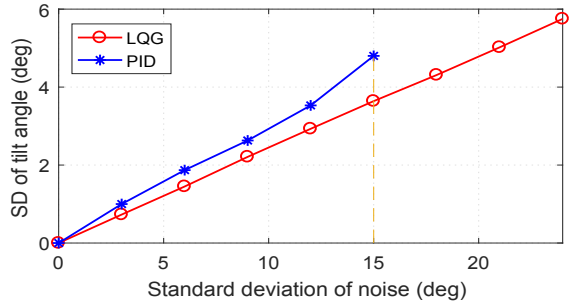

Fig. 9. The stability of system when increasing noise amplitude

angle versus the standard deviation of the measurement noise. It can be seen that LQG is able to maintain system stability better than PID when increasing the amplitude of noise. As such, the LQG shows the superior performance comparing to PID in CMG application.

B. Experimental results

In the experiment setup, two angular position sensors are used for measurement of the tilt angle and gimbal angle. The hardware configuration model shown in Figure 10. The microcontroller reads the angular positions of θ and α as input signals of control algorithm. The controller calculated corresponding control signal which is then fed into driver block of gimbal motor. The driver block converts control signal to control voltage and then manipulate gimbal motor. This process repeates continuously to balance the system. The control algorithm is processed by microcontroller STM32F407. The CMG model has the same parameters as Table I except the maximum angular velocity of flywheel is $\omega = 9000$ rpm. Due to the limitation of applying PID for this system as demonstrated in simulation, only LQG algorithm is implemented in this experiment.

The pictures of CMG are shown in Figure 11. When the control system is turned off, the CMG is tilted at $-30°$ as illustrated in Figure 11(a). By performing LQG, the CMG is stabilizing around equilibrium point as shown in Figure 11(b).

Figure 12 shows the behavior of the experimental system in terms of tilt angle θ and gimbal angle α over time, that the CMG stabilized the inverted pendulum from the tilt angle

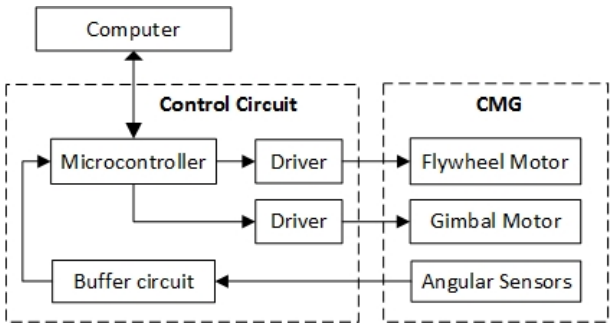

Fig. 10. Hardware configuration model

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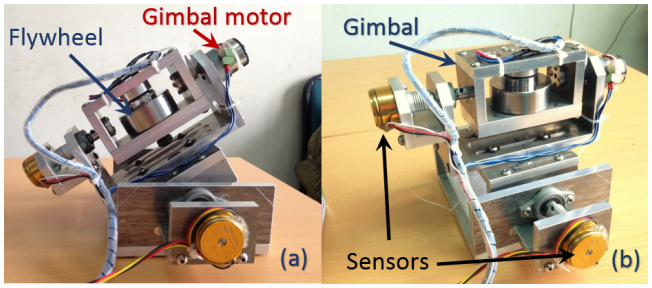

Fig. 11. The experiment setup: (a) when CMG does not work; (b) when CMG is stabilizing

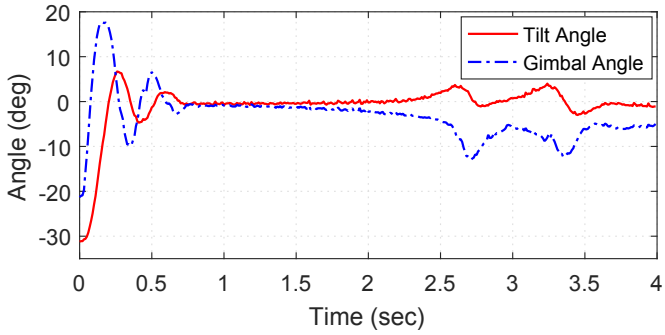

Fig. 12. Experiment result with LQG regulator

of $-30°$ in less than one second with a maximum overshoot of $6°$. At 2.5 seconds, the inverted pendulum is tilted by a small angle. This phenomenon due to the characteristics of the gimbal motor which is realized using a DC motor. When the tilt angle is small, the controller responds to the small tilt angle by providing a small voltage. However, the DC motor does not operate when the absolute value of control voltage is less than a certain value, this voltage range is called deadzone [16]. Therefore, for small tilt angle, the CMG do not generate the torque for the gimbal to counteract the falling. Nevertheless, in the following moment, as the tilt angle is large enough, the controller generates high enough voltage for the DC motor to overcome this situation and keep system balance.

In the next experiment, the CMG's balancing ability is tested under external force. Figure 13 shows tilt angle θ and gimbal angle α over time. At first, the system is balanced as tilt angle is very small. At 1.9 seconds, the system is suddenly affected by an external force. Shortly after, CMG controls the gimbal to balance the system in less than 1 second. It demonstrates that the CMG responses rapidly to ensure the system stability.

V. CONCLUSIONS

In this study, the auto-balancing algorithms of the static object using CMG are tested by both simulation and experiment. The CMG employing LQG regulator performs better than that of employing PID one. With LQG, the CMG can stabilize an object with weight up to 15 times more than that of the flywheel. The algorithm is also designed to simultaneous optimizing the tilt angle and gimbal angle. This

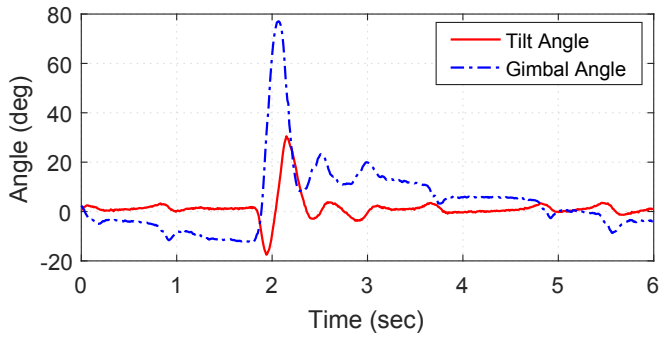

Fig. 13. System response when system affected by external forces

ensures system stability even when it is affected by multiple of external disturbance. Future controller can be designed by combining controller with a compensatory algorithm to overcome dead-zone of DC motor. This can helps the system avoid lost-balance state and might increase the system's ability to counteract the external forces.

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